

Measurements of Optical Tissue Properties Using Integrating Sphere Technique

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Abstract

The optical properties of white matter human brain, canine prostate and pig liver were measured in the wavelength range 330-1100 nm. The measurements were carried out in native as well as in coagulated tissues. We used the double integrating sphere technique to provide reflection and transmission measurements and a special homogenising technique to prepare the tissue. The optical properties were evaluated using an inverse Monte-Carlo simulation, considering the geometry of the experimental set-up. All tissues show characteristic absorption bands at 420 nm and 550 nm, related to the strong absorption of haemoglobin. After coagulation the scattering increases drastically while absorption remains nearly unchanged. The anisotropy factor g increases with increasing wavelength and drops down slightly after coagulation. The wavelength behaviour of tissue scattering has been compared with theoretical calculations (Mie-theory), showing that ideal spheres with an diameter between 0.6 and 0.8 μm fit best to the experimental results.

1. Introduction

To get a deeper understanding of most laser applications in medicine the knowledge of the optical properties of the irradiated tissue is indispensable. This is especially valid for dosimetrical questions. Because the tissue properties often change during the laser treatment, their temperature behaviour must be known as well¹.

The propagation of laser light in biological tissue and the transformation into thermal energy due to absorption of photons is governed by the optical tissue properties. They are given by scattering from inhomogenities (membranes, nuclei, etc.) and absorption in constituents (NADH, haemoglobin, melanin, water, etc.). The corresponding physical parameters are the scattering coefficient μ_s [mm^{-1}] and the absorption coefficient μ_a [mm^{-1}], both being wavelength dependent.

For completeness, the scattering phase function $p(s,s')$ must be known. It describes the probability of a photon to be scattered from a direction s to a direction s' . Since the phase function is the only parameter which can be measured independently using thin slices and goniometrical equipment, several functions have been discussed in the literature to enable the best fit to experimental data (isotropic, Henyey-Greenstein, Delta-Eddington, Rayleigh-Gans, etc.)^{3,5,6,7,10}. From these, the Henyey-Greenstein function is the one which is most often used in tissue optics because of its mathematical simplicity and its good correspondence to experimental data. With a given phase function, the scattering behaviour can be described by the anisotropy factor g which can vary from -1 up to +1. In our Monte-Carlo calculations we used the Henyey-Greenstein phase function in order to determine g from experimental data (Θ scattering angle):

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$$p_{HG}(\Theta) = \frac{1 - g^2}{(1 + g^2 - 2g \cos \Theta)^{\frac{3}{2}}}$$

$$g = \int_0^{2\pi} d\Phi \int_0^{\pi} p(\Theta) \cos \Theta \sin \Theta d\Theta$$

One method to measure the complete set of optical properties with only one experiment is the double integrating sphere technique^{7,11,20,22,23}. With collimated illumination, the measurements of the diffuse reflectance (R_{cd}), the diffuse transmittance (T_{cd}), the collimated (unscattered) transmittance (T_c), and the thickness of a sample (d) permits the complete determination of its optical properties using a model of radiative transfer (fig. 1). One has to distinguish between collimated and diffuse illumination in the nomenclature: while the index 'cd' will be used for collimated illumination, the index 'd' corresponds to diffuse illumination.

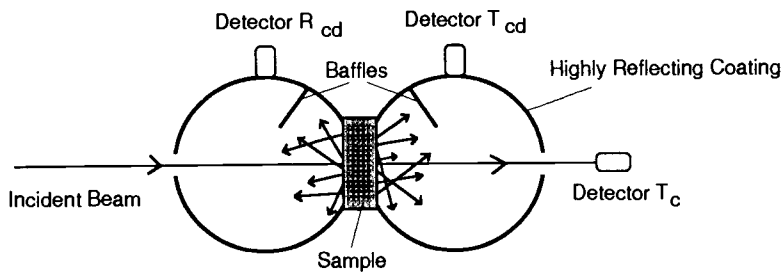


Fig 1: Double integrating sphere geometry for the measurement of R_{cd} , T_{cd} , T_c

In advance of calculating μ_a , μ_s and g from reflection and transmission measurements with integrating spheres, the "true" values for R_{cd} and T_{cd} have to be determined since they are slightly different from the measured ones which are indicated with an asterisk (R_{cd}^* , T_{cd}^*). These corrections are necessary because of the exchange of light through the sample between both spheres. Moreover, the relative position of the inner shield (baffle) of the transmission sphere has an influence on the measuring values (the position is different for calibration with reflecting standard and sample measurements). A complete description is given by Pickering¹¹.

Several models are described in the literature to calculate optical properties from experimental data⁷:

1. Kubelka-Munk-Theory can only deal with diffuse irradiation and is limited to those cases, where scattering dominates absorption ($\mu_a < \mu_s(1-g)$)^{2,5,9,14,15}.
2. The diffusion approximation of the photon transport equation is not restricted to diffuse irradiation but is also limited to those cases where reduced scattering ($\mu_s' = \mu_s(1-g)$) exceeds absorption significantly^{2,4,5,9}. Since scattering in biological tissue is highly anisotropically, both models apply only for optical thick layers ($d \gg (\mu_a + \mu_s)^{-1}$). The consideration of boundary conditions as given in our experiment is difficult and an approximation is given by Keijtzer¹⁶.

3. The adding-doubling method can be used for every set of optical properties and boundary conditions^{3,17,23,26}. From all one-dimensional models, this method provides the most accurate results, if the diameter of illumination is small compared to the sample diameter.
4. Monte-Carlo-Simulations yield the most exact solutions since every geometric parameter can be considered in the program. The main disadvantage is the inverse calculation algorithm using optical properties to calculate reflection and transmissions. Thus the optical parameters have to be fitted to the measured values, requiring long computation times^{18,19,20,21,27}.

Regarding the integrating sphere technique the sample diameter cannot be assumed to be much more extended than the diameter of the illuminating spot. In our arrangement this ratio is 13:5, far away to be infinite as a priori assumption for an one dimensional description. The assumption of infinite sample extension is badly met in the NIR due to the large penetration depth of the photons, thus we decided to use an inverse Monte-Carlo-Simulation to calculate optical properties. In this wavelength range one dimensional models (e.g. Adding-Doubling) yield virtually increased absorption coefficients because of misredarded sideway losses of photons. In spite of this, we used the Kubelka-Munk formulas to provide fast approximations to the optical parameters which were used as start parameters for the Monte-Carlo simulation.

2. Experiment

The measurements were carried out with two integrating spheres (Labsphere Inc., $\varnothing=30\text{cm}$) and intervening samples of different tissues. The output of a tuneable light source (Xenon lamp with monochromator) was mechanically chopped with 600 Hz and focused at the sample, placed inside a glass-cuvette with calibrated thickness. The spot size was approx. 5 mm in diameter and the beam divergence was approx. 1.2 deg. We adjusted the slits of the monochromator to achieve a FWHM of 10 nm. The measurements were carried out in the range from 330 nm to 1100 nm, restricted by the sensitivity of the photodiodes. The signals were amplified using lock-in technique and recorded by a personal computer.

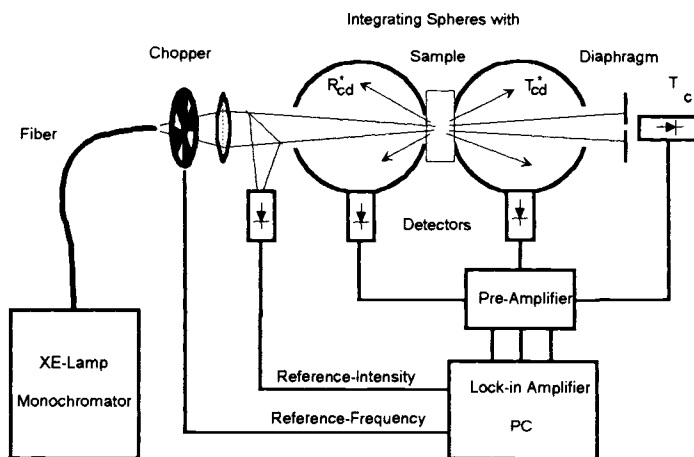


Fig. 2: Experimental set-up used for the measurements of optical tissue properties

The measurements for each sample were carried out in three steps for all wavelengths of interest, both for the native as for the coagulated tissue state:

1. In advance of each measurement the system was calibrated without sample using a reflection standard (99%) for both sphere intensities and the center part intensity of the free light beam for the collimated intensity. For all measurements the influences of parasitic light and intensity fluctuations of the source were taken into account by normalisation to the reference intensity.
2. The sample was placed between the spheres using special cuvettes (see tissue preparation). While the specular reflectance was leaving the first sphere without being detected, the diffuse reflectance (R_{cd}^*) and the total (diffuse and collimated) transmittance ($T_{cd}^* + T_c$) were measured relatively to the reflection standard by silicon photodiodes placed in the walls of the spheres. For these measurements the second sphere was closed at its exit hole by a reflecting baffle. There are two reasons to measure the sum of diffuse and collimated transmittance in the second sphere:
Firstly, many of the photons scattered only once or twice leave the sample within the angle of divergence of the incident beam because of the highly forward peaked tissue scattering ($0.9 < g < 1$ in the NIR). These photons could leave the second sphere without being detected as diffuse transmittance if no closing reflecting baffle is used. Secondly, the calibration of the second sphere after removing the reflecting standard is no longer valid because of higher internal losses.
3. To measure the collimated transmission, the baffle was removed from the second sphere and the intensity of the center part of the transmitted beam was detected behind a diaphragm ($\varnothing$: 1mm, distance to sample: 650 mm). The diffuse transmittance was finally obtained by subtracting the collimated transmittance from the total transmittance measured in the second sphere.

2. Tissue preparation

The tissue was prepared after resection (0.5-3 hours post mortem) by cooling it in liquid nitrogen and pulverising it in a mortar. This procedure provides cell clusters of 50-500 μ m which are larger than one cell (approx. 5 μ m). Since the sample was frozen very fast, no ice crystals could appear, thus leaving the cell membranes undestroyed. Doing this, we assume that the homogenised tissue has the same optical properties as the initial tissue. This was also shown by Peters et al.²², who compared the optical properties of samples prepared by this method with samples which were cut by a microtom. The main advantage of this procedure is the knowledge of the exact sample thickness in combination with calibrated cuvettes since it is difficult to achieve exact frozen cuts in the range between 60 μ m and 500 μ m with standard microtom technique. We used cuvettes with thicknesses of 100, 200 or 500 μ m.

After measuring a sample it was left in its cuvette and coagulated in a bath of water at temperatures of approximately 75°C. After this, the sample was measured again, providing the thermally induced change of the optical properties. We assume that the thermal effects in a bath of hot water are nearly the same as those during laser coagulation, if the temperature does not exceed 100°C.

3. Calculation of the optical properties from experimental data

Correction of the measured values due to the sphere equations:

As mentioned in the introduction, the measured values for R_{cd}^* and $T_{cd}^* - T_c$ have to be corrected for the exchange of light between both spheres and for a change of the sphere losses with changing sample remission. The presented formulas are only valid for our arrangement of integrating spheres shown in fig. 1, including the baffle positions, the calibration procedure and the measuring procedure. The correct values for R_{cd} and T_{cd} can only be calculated from R_{cd}^* and T_{cd}^* with an iteration since they depend on each other. Thus some assumptions must be made for the first iteration step, depending on the reflection of the calibrated standard (R_{99}):

$$R_{cd,n=1} = R_{cd}^* \cdot R_{99} \quad T_{cd,n=1} = (T_{cd}^* + T_c) \cdot R_{99} - T_c$$

According to the nomenclature of Pickering^{11,20}, in the following formulas primes indicate values for the transmission sphere while no primes accord to the reflectance sphere. After some calculations one finds:

$$R_{cd,n+1} = \frac{R_{cd}^* R_{99} \left\{ 1 - \left(m\alpha + R_{d,n} \frac{s}{A} \right) \right\} (1 - T_n T_n')}{1 - \left(m\alpha + R_{99} \frac{s}{A} \right)} - T_n' (T_{cd,n} - mT_c)$$

$$T_{cd,n+1} = \frac{1}{m\alpha'} \left[\frac{(T_{cd}^* + T_c) R_{99} \left\{ 1 - \left(m\alpha' + R_{d,n} \frac{s}{A'} \right) \right\}}{1 - m\alpha'} - T_n R_{cd,n} m\alpha' - mT_c \right]$$

with

$$T_n = \frac{sT_{d,n}}{A \left\{ 1 - \left(m\alpha + R_{d,n} \frac{s}{A} \right) \right\}} \quad T_n' = \frac{sT_{d,n}}{A' \left\{ 1 - \left(m\alpha' + R_{d,n} \frac{s}{A'} \right) \right\}}$$

$R_{cd}^*, T_{cd}^* + T_c$	Measured values, relative to the calibrated standard
R_{99}	Reflection factor of calibrated standard
R_{cd}	Reflection of sample for collimated irradiation
R_d	Reflection of sample for diffuse irradiation
T_{cd}	Transmission of sample for collimated irradiation
T_d	Transmission of sample for diffuse irradiation
T_c	Transmission of sample for unscattered photons
m	Reflection factor of the sphere walls
A, A'	Total sphere area
d, d'	Area of detector
s	Area of sample
h, h'	Area of all holes
α	$1 - (d/A + s/A + h/A)$
α'	$1 - (d'/A' + s/A' + h'/A')$

Within the equations some values for diffuse illumination appear (R_d, T_d) since the 'diffuse' radiation field inside the integrating spheres acts as an additional light source. Unfortunately, they are unknown for the sample. We decided to use approximations in the iteration procedure, depending on the actual values for collimated illumination.

$$\begin{aligned}R_{d,n} &\approx R_{d,glass} + (1 - R_{d,glass})R_{cd,n} \\T_{d,n} &\approx T_{d,glass}(T_{cd,n} + T_c) \\R_{d,99} &= R_{cd,99}\end{aligned}$$

The approximations are based on the assumption that the reflection for diffuse illumination is larger than for collimated illumination due to the higher Fresnel reflection at the enclosing glass slides. With the same argument the transmission decreases for diffuse illumination. The values for the reflection and transmission of the air/glass interface for diffuse illumination were determined numerically from the Fresnel equations. For $n_{glass}=1.45$ the calculation yields $R_{d,glass}=0.2$. The transmission was approximated by $T_{d,glass}=1-R_{d,glass}$. The iteration to find R_{cd} and T_{cd} was carried out by the computer in advance of running the Monte-Carlo simulation. For most cases it converged within our accuracy for less than 10 cycles.

Approximate evaluation of the optical properties using Kubelka-Munk theory:

To get a rough and fast approximation for the measured optical parameters, we used the simple Kubelka-Munk formulas. As mentioned, they can only deal with diffuse illumination without any boundary conditions and the assumption is made, that scattering dominates absorption. In spite of this, for most sets of parameters the differences to the Monte-Carlo-Simulation were within 100% for μ_a and within 20% for μ_s and g .

Firstly, we corrected the measured values to eliminate the interaction of the two integrating spheres as mentioned above. Then we changed the values slightly to consider the boundary conditions in a rough approximation (correction due to specular reflection):

$$T'_c = 1.08 \cdot T_c \quad R = 1.04 \cdot R_{cd} \quad T = 1.04 \cdot T_{cd} + T'_c$$

Now we applied the well known Kubelka-Munk equations:

$$\mu_a = \frac{A}{2} \quad \mu_s = \left(\frac{-\ln T'_c}{d} \right) - \mu_a \quad g = 1 - \frac{(\mu_a + 4S)}{3\mu_s}$$

with

$$\begin{aligned}S &= \frac{1}{bd} \ln \left(\frac{1 - R(a-b)}{T} \right) & A &= (a-1)S \\ a &= \frac{1 + R^2 - T^2}{2R} & b &= \sqrt{a^2 - 1}\end{aligned}$$

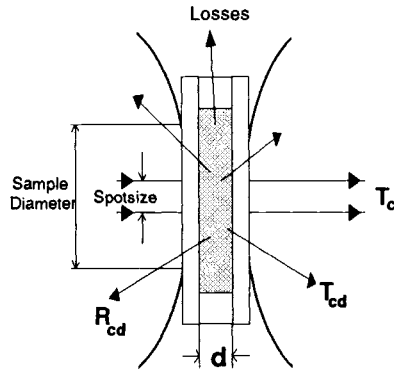


Fig. 3: Collimated irradiation of a tissue sample, placed inside a glass-cuvette

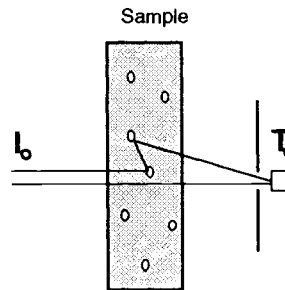


Fig. 4: Mismeasurement of collimated transmission due to parts of diffuse transmission

Evaluation of the optical parameters using an inverse Monte-Carlo simulation:

As can be seen from fig. 3 and as mentioned in the introduction, a model to calculate optical parameters from experimental data has to consider boundary conditions given by the environment (glass, air). Besides this, at some wavelengths the consideration of geometrical parameters (sample diameter, spot size, etc.) is very important to calculate losses of photons which are not caused by absorption or to consider parts of the measured values for the collimated transmission which are caused by diffuse transmission (fig. 4). If one-dimensional models (e.g. adding-doubling, Kubelka-Munk) are used to evaluate optical properties from transmission and reflection measurements, the consideration of these effects can only be carried out in a rough approximation²³.

To describe the 3-dimensional arrangement we decided to use a Monte-Carlo technique. The main disadvantage of relatively large computational expenditure can be balanced by high performance PC-machines (486/CPU) with an external mathematical coprocessor. To achieve a practicable simulation speed, we produced assembler like software performance. Thus we preferred to use a language well established in solving scientific and technical problems: FORTRAN (now available as FORTRAN90 by LAHEY or SALFORD).

Fig. 5 shows a general description of the software for calculating the penetration of a single photon, considering the total experimental set-up. Accumulating a sufficient number of photons stabilises the results for backward, forward and collimated intensities. Starting with appropriate initialising parameters from the Kubelka-Munk equations, we constructed a derivation matrix for the individual intensities by changing every optical parameter slightly step by step. For every step the number of photons was chosen sufficiently large to produce stable results as a necessary presupposition to construct a smooth plane in the phase space. In this way the three measured intensities can be approached due to Fig. 6 by the conventional gradient method.

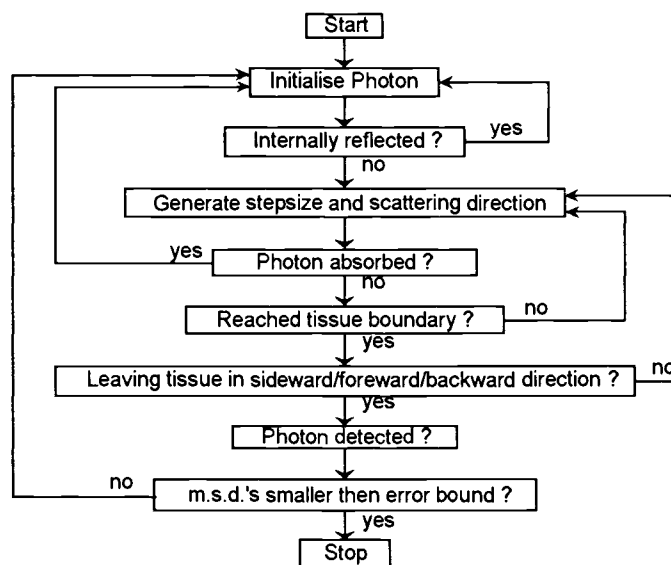


Fig. 5: Flowchart of the Monte Carlo technique

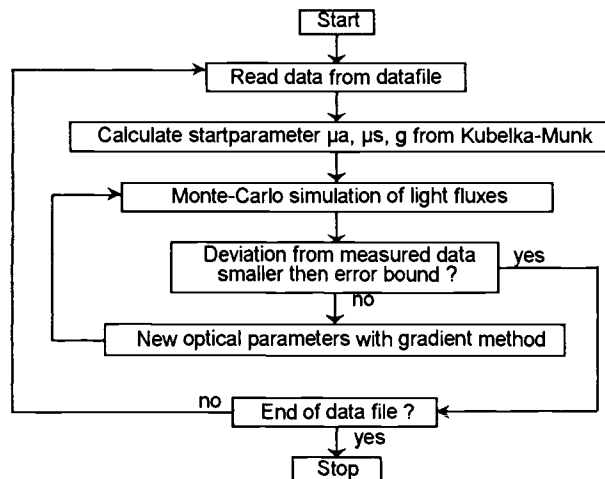


Fig. 6: Non-linear approximation software using gradient method

The iterative character of the procedure is caused by the impossibility to describe measured intensities analytically, faced with a complex geometry (mismatched boundaries, detector arrangement, etc.) and with a common wide range of the albedo and the optical thickness of the sample. Thus there are no forward equations but only forward simulations. Consequently, the inverse problem of extracting optical parameters from measured intensities can be solved only by using a numerical optimising method. Regardless the complexity of the problem, there are some fundamental restrictions in constructing derivations in a useful way. For example the increase of absorption leads to a decrease in all detected

intensities. Our simulations confirm this consideration for all changes in μ_a , μ_s and g . In general, these considerations provide a schedule due to Fig. 7 allowing to decide whether an element of the derivation matrix can be accepted or not. A mismatch in sign indicates strongly overwhelming statistical impact, leading to an automatical increase of the number of simulated photons.

$$\text{sign} \begin{pmatrix} \frac{\Delta I_r}{\Delta \mu_a} & \frac{\Delta I_r}{\Delta \mu_s} & \frac{\Delta I_r}{\Delta g} \\ \frac{\Delta I_f}{\Delta \mu_a} & \frac{\Delta I_f}{\Delta \mu_s} & \frac{\Delta I_f}{\Delta g} \\ \frac{\Delta I_d}{\Delta \mu_a} & \frac{\Delta I_d}{\Delta \mu_s} & \frac{\Delta I_d}{\Delta g} \end{pmatrix} = \begin{pmatrix} -1 & +1 & -1 \\ -1 & -1 & +1 \\ -1 & -1 & +1 \end{pmatrix}$$

Fig. 7: General schedule of signs of the derivation matrix

Furthermore, the procedure can be speeded up by taking into account aspects of self-learning software. Thus we stored the results of every successful simulation. In advance of starting a new simulation with changed parameters, the software will check if a set of similar parameters already exists from past simulations. Thus one can avoid similar simulations by building up an archive for the used experimental set-up. The self-learning environment will influence the time performance of the software more and more.

Illustrating further explanations, fig. 8 shows the results of Monte-Carlo simulations for some human tissues according to sideways losses and absorbed intensities. As stated above, sideways losses can be of the same amount as absorbed intensities. Ignoring their influence, the absorption coefficient of the sample would be virtually increased. This indicates the limit of a usual slab geometry software.

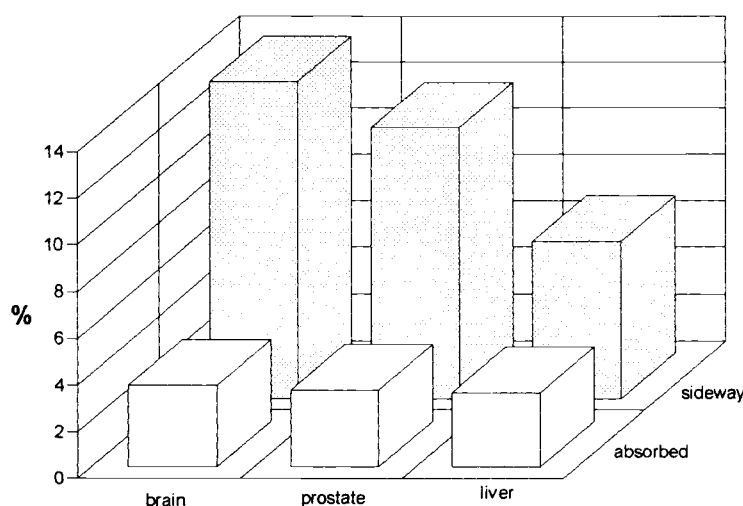


Fig. 8: Sideway losses and absorbed intensities of different native tissues at 1064nm, calculated with Monte-Carlo software. The optical properties according to table 1, the thickness of the samples was 0.5 mm.

As shown in fig. 9 both, the collimated and the diffuse intensities at the detector for collimated transmission show a constant behaviour in a wide range of the detector diameter. This indicates two important features about the experimental set-up: Using a relatively small beam divergence, the collimated intensity is constant near the optical axis in the detector plane. Of course one can verify this fact by analytical considerations. More basically the diffuse intensity shows the same behaviour in the wide range of experimentally used sample thicknesses. The results show the same behaviour for strong scattering (coagulated) tissues.

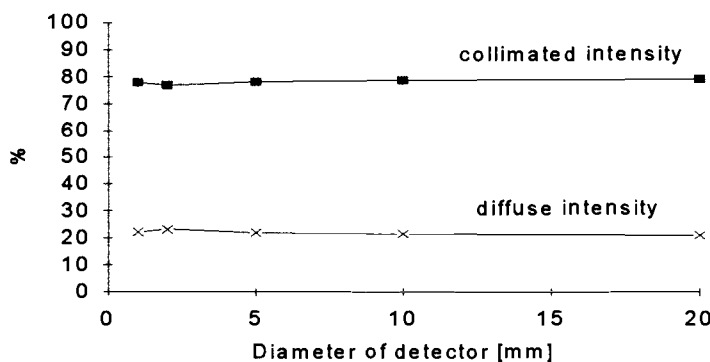


Fig. 9: Different parts of the measured intensities at the detector for the collimated transmission (according to fig. 4) of native pig liver at 800nm for 0.2 mm thickness, calculated with Monte Carlo software. The spot size at the sample was 5 mm, the distance between sample and detector plane was 650 mm and the spot size in the detector plane was 40 mm (i.e. 1.54° beam divergence). Intensities are normalised to the sum of the total detected intensity at the diode for T_c .

Choosing a relatively large diameter for the detector of the collimated transmission will suppress noise impact in experimental measurements. Besides this, it influences the analysing software by rapid convergence of the relative intensities. One can imagine, that the diffuse illumination of the collimated detector cannot be neglected generally.

4. Results

Measurements were carried out for different types of tissues (white matter human brain, canine prostate and pig liver) in the range 330-1100nm with an increment of 25 or 50 nm. Because of the excessive computation time for the inverse Monte-Carlo Simulation, we first present the complete spectra calculated with Kubelka-Munk theory (fig. 12-14). Here we give no information according to the mistakes since the curves should only give a rough impression of the wavelength dependence of the optical properties. Additionally the calculations were carried out with the Monte-Carlo method for some wavelengths in the NIR, since the penetration depth of photons has a maximum for most tissues in this range, making the NIR very interesting for optical tomography.

		human brain			canine prostate			pig liver		
		μ_a [mm ⁻¹]	μ_s [mm ⁻¹]	g	μ_a [mm ⁻¹]	μ_s [mm ⁻¹]	g	μ_a [mm ⁻¹]	μ_s [mm ⁻¹]	g
800 nm	native	0.08	14	0,95	0,06	13	0,96	0,14	10	0,96
	coag.	0.09	17	0,94	0,07	27	0,95	0,15	45	0,94
1064 nm	native	0.04	11	0,95	0,04	11	0,96	0,05	8	0,97
	coag.	0.05	13	0,93	0,05	22	0,96	0,06	30	0,95

Tab. 1: Optical properties of some tissues in the NIR. Results with inverse Monte Carlo Simulation. The mistakes are assumed to be $\pm 20\%$ for μ_a , $\pm 10\%$ for μ_s and $\pm 2\%$ for g, given by the statistic of several measurements.

The results are in good agreement with experimental data presented by other authors using complex models to calculate their data^{20,23}, considering that they made some assumptions on the g-factor (e.g. fixed to 0.9) which influenced their results for μ_s or μ'_s . As can be seen from table 1 the absorption does not change drastically during coagulation, while scattering shows a strong increase. In the NIR the g-factor remains nearly unchanged for different wavelength and is independent from the tissue state. The spectra of the absorption coefficients (fig. 12-14) show absorption peaks at 420 and 550 nm which we relate to the absorption bands of haemoglobin²⁴. The absorption coefficients for the coagulated tissues calculated with Kubelka-Munk theory were not shown, since they are virtually increased in this model. We made the assumption, that they remain nearly unchanged for a wide range of wavelength.

The spectra of the scattering coefficients show a smooth decrease over the measured wavelength range. This corresponds well with the predictions of Mie-theory²⁵. We made some calculations in comparing the dependence of tissue scattering on the wavelength with theoretical results from Mie-theory. Assuming ideal scattering spheres (radius r , index of refraction $n=1.6$) in water ($n_0=1.33$), their wavelength dependent scattering cross section was calculated numerically²⁸:

$$\sigma_s = \frac{2\pi}{\lambda} \int_{4\pi} p(\Theta, \Phi) d\Omega$$

Here λ is the wavelength and $p(\Theta, \Phi)$ is the Mie-phasefunction which depends on the Mie parameters:

$$m = \frac{n}{n_0} \quad \text{and} \quad x = \frac{2\pi r}{\lambda}$$

The scattering coefficient μ_s may be calculated for different concentrations by multiplying σ_s with the particle density. The anisotropy factor g was determined by numerical solvating of the corresponding integral which is described in the introduction. The results for scattering spheres with a radius of 0.3 μm in the wavelength range 380-660 nm are shown in fig. 10.

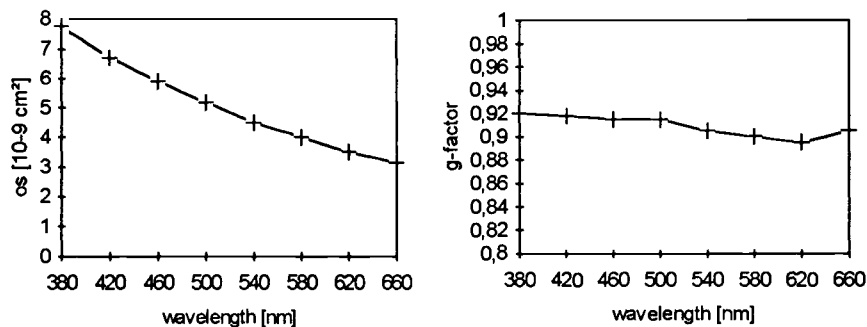


Fig. 10: Scattering cross section and g-factor of ideal spheres ($n=1.6$, $n_0=1.33$, $r=300\text{nm}$), calculated with Mie theory

To compare the theoretical results with our experimental data, we decided to calibrate the particle density of the virtual suspension in such a way that the resulting scattering coefficient fits to the tissue scattering of native human brain at 500 nm. Comparing the wavelength dependencies of the scattering coefficients for spheres with different radii, the curves for $r=300\text{nm}$ and $r=400\text{nm}$ fit best to the measured tissue behaviour (fig.11). This holds also for the g-factor in a wide range of wavelength as can be seen from the comparison of fig. 10 and 12.

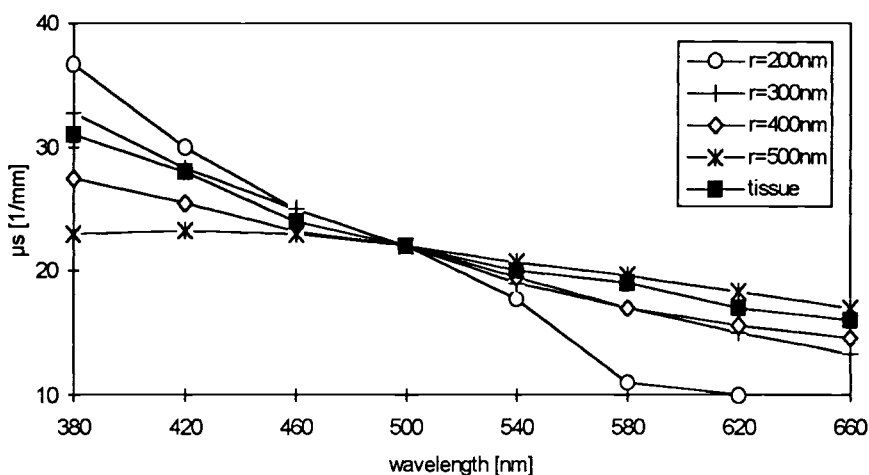


Fig. 11: Wavelength dependence of the scattering coefficient of ideal spheres with different radii and measured values for native pig liver.

The calculations show in a rough approximation, that the scattering centers in biological tissues are smaller than the cells itself. This gives an additional reason for our method of tissue preparation since scattering structures are sub-cellular. From our calculations the particle density of scatterers for $r=300 \text{ nm}$ is approx. $40 \cdot 10^9 \text{ cm}^{-3}$, thus filling approx. 0.5% of the total tissue volume.

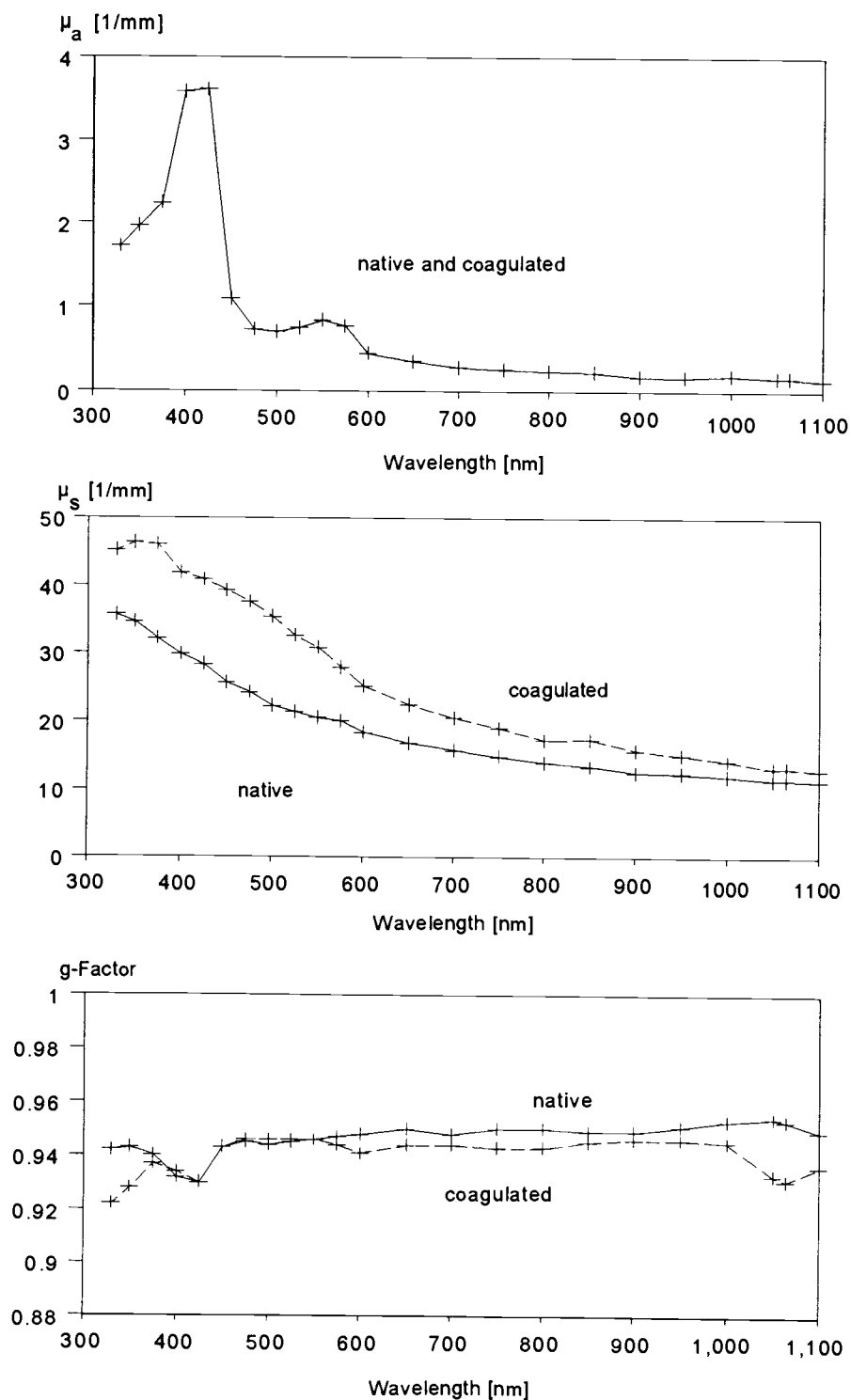


Fig. 12: Optical properties of white matter human brain, calculated with Kubelka-Munk theory

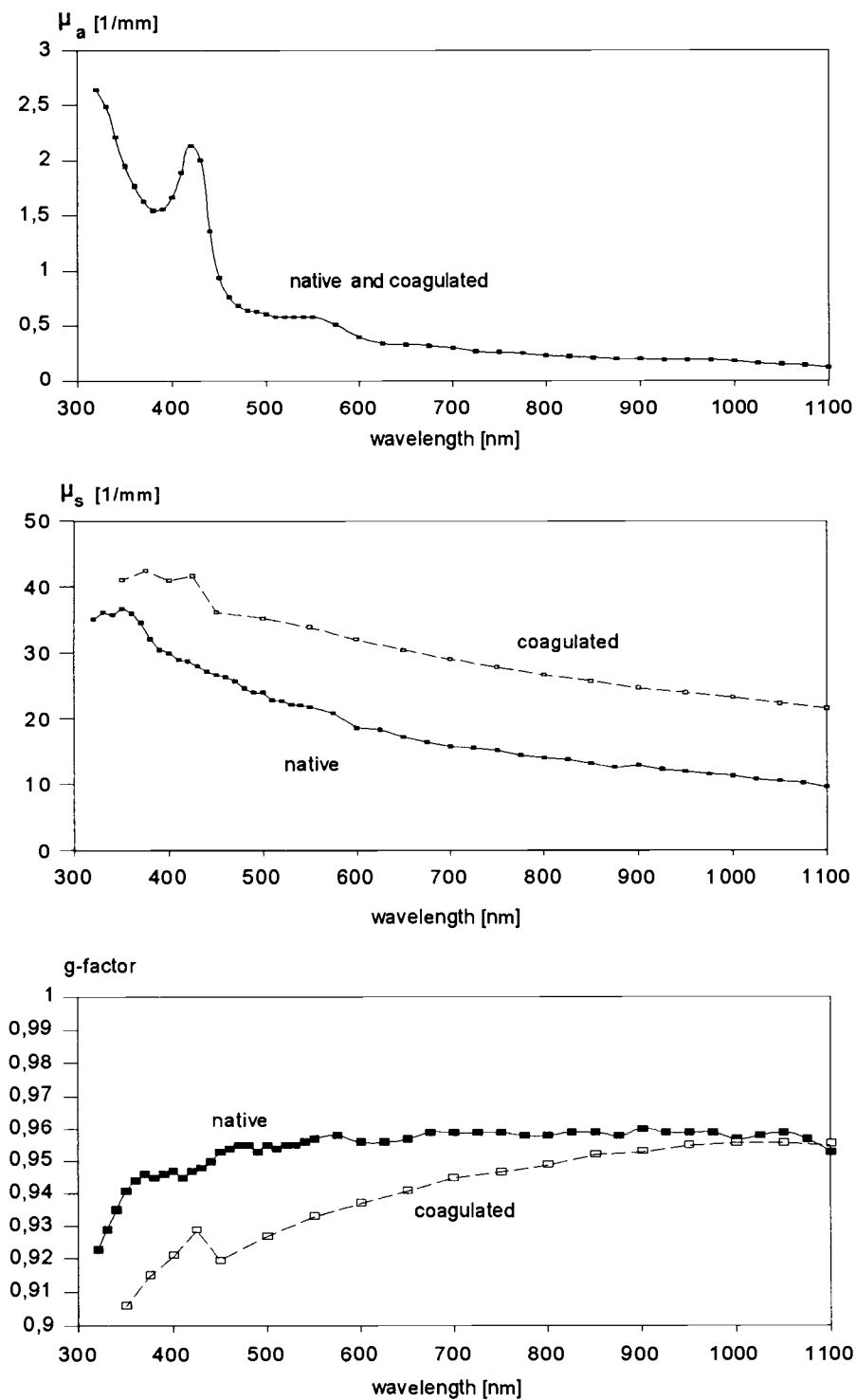


Fig. 13: Optical properties of canine prostrate, calculated with Kubelka-Munk theory

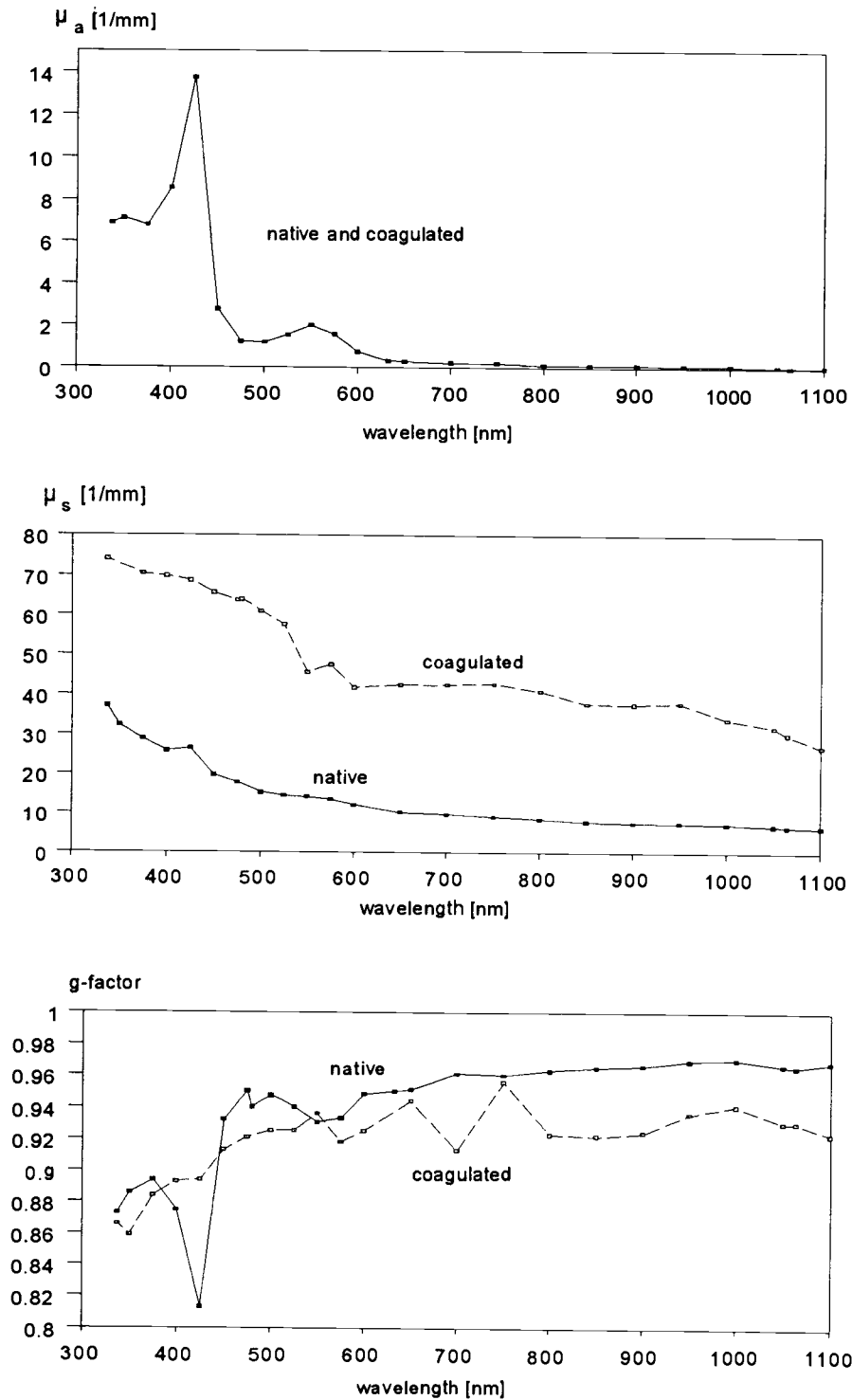


Fig. 14: Optical parameters of pig liver, calculated with Kubelka-Munk theory

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